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FACHBEREICH ELEKTROTECHNIK
UND INFORMATIONSTECHNIK

Power System Operation - Lab

Prof. Dr. Sebastian Weck, Lars Weispfenning

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Session 2

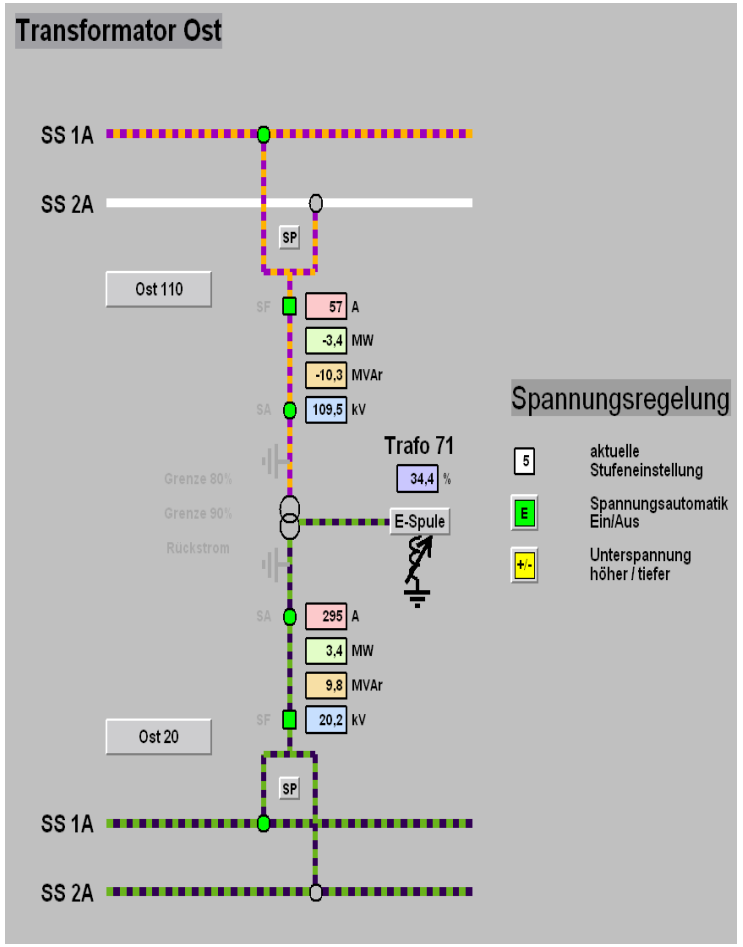
Exercise 6-8

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Exercise 6

Fast replacement of a defective circuit breaker (CB) and restoration of mains protection

- In the OST 110 plant, the actuator of the 110kV circuit breaker (CB) of transformer 71 has gone defective in the ON state, i.e. it can no longer be switched OFF. (CB can be blocked by trainer)
- For switching and in case of short circuit the CB could not switch OFF any more.
- Consider how to provide selective protection using the bus bar cross-coupler on 110 kV.
- What protection settings must the bus bar cross-coupling have?



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Exercise 7

Voltage and load control at the MV transformer

- The load voltages of the MV and LV customers are adjusted at the MV transformers (TR69, TR70, TR71, TR72) by longitudinal steps (+/-13). Typically, there is a voltage regulator. Regulator "ON" means automatic stepping to the target voltage 20.5 kV +/- about 400 V. i.e. about +/- 5V to 230V.
- Switch the mains so that TR69 or TR70 alone carries the mains group West20.
- Switch the voltage regulator to " OFF".
- Manually change the level by a few +/-stages and observe and note the consequences:
  - MV voltage at the busbar
  - The active and reactive power of the load
  - How are the loads voltage dependent?

Stage	U [kV]	P [MW]	Q [MVAr]

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Exercise 8

Planned network switchovers for transformer replacement feed-in

- Put yourself in the following scenario: The TR 71 transformer reports "Gas formation stage 1". An insulation fault is imminent. TR 71 is to be shut down as soon as possible, the oil quality is to be checked, etc....
- TR 71 must be replaced; the customers of the connected network group must continue to be supplied without interruption. Lines K1 and K2 to wind 20 are not available, please switch off!
- Plan the disconnection of TR71: Which services are missing in case of disconnection? Estimate: What loads (%) will occur during switchover?
- Use a transformer in substation “West 20” as a substitute and switch a supply from “West 20” via “Mitte 20” or/and “Süd 20” in such a way that transformer 71 is replaced via the grid lines. Keep in mind the maximum currents of the lines (400A) and the expected voltage of the remote bus bar in “Ost 20”.
- Do not switch off TR 71 until you are sure that the changeover will take place without a fault